Habiba Morsy

CONTACT

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Profiles: Linkedin, Github, Website

TEACHING EXPERIENCE

Graduate Teaching Assistant
University of Virginia

- Courses:
  - Data Eng II: Big Data Systems: Master's Level — **08/2026 – 12/2026**
  - Machine Learning in Systems and Network Security: Master's Level — **06/2026 – 08/2026**
  - Introduction to Deep Learning: Undergraduate Level — **01/2026 – 05/2026**
  - Computational Probability: Undergraduate Level — **09/2025 – 12/2025**

Student Assistant — **09/2023 – 05/2024**
Kean University

- Tutored students in Python and Java programming concepts and debugging techniques.
- Simplified data structures, algorithms, and OOP principles for diverse learning needs.

WORK EXPERIENCE

Software Developer Intern — **05/2024 – 08/2024**
RCSB Protein Data Bank

- Developed and implemented a Python wrapper for RCSB PDB's data API to facilitate streamlined access and integration of 3D macromolecular structure data
- Contributed to the open-source community by publishing the Python package on GitHub and PyPI
- ·Tools and libraries used: GraphQl, rustworkx, networkx, Microsoft Azure, Github, PyPI
- ·Github Repository, PyPI Package

RESEARCH EXPERIENCE

Physics-Aware Deep Learning for Aerodynamic Design — **06/2026 – Present**
University of Virginia

- Developing an automated Formula 1 geometry-to-CFD pipeline that generates aerodynamic datasets for physics-aware surrogate models to predict aerodynamic performance across unseen designs.

Beyond Idealized Voids: Linking Pore Morphology to Hotspot Sensitivity — **07/2026 – Present**
University of Virginia

- Developed a Statistical Shape Model to create an interpretable latent design space for systematically generating realistic microstructures and linking morphology variations to shock-induced hotspot sensitivity using LatentPARC.

Independent Research — **09/2023 – 05/2024**
American Council on Education & Kean University

- Developed an LLM- and RAG-powered application for CCIHE queries, dynamically updating the Q&A database and integrating with a React dashboard for an enhanced user experience.

Undergraduate Research Assistant **01/2022 – 05/2023**
Kean University, Advisor: Dr. David Joiner, Acting Associate Dean

- Developed interactive VR applications in Unity, implementing complex data interactions and user interfaces while creating and optimizing 3D models using Blender and Maya.

EDUCATION

University of Virginia **08/ 2025 - 05/ 2030**
Doctor of Philosophy - PhD in Data Science

Kean University
M.S Computational Science and Engineering **01/ 2025 - 06/2025**
(one semester - transferred to UVA)

- Master's Thesis: Developing Large Language Models to Automate Quality Assurance of Digital Library Metadata

B.S. in Science and Technology **09/ 2021 - 12/ 2024**
Applied Mathematics Track, Computer Science Minor

- Dean's List 2021-2025
- Vice President, Association for Computing Machinery
- Treasurer, Association for Computing Machinery - Women
- Secretary, Arab Culture Club

AWARDS

- Generation Google Scholar
- National Cyber Scholarship Scholar
- NJ Seal of Biliteracy, French, English, Arabic
- Distinction in Research Award - Kean University
- STEM Scholarship Recepient

PUBLICATION

RCSB Protein Data Bank: Python Software Package for Accessing RCSB PDB APIs

- Journal of Molecular Biology

HPC-ED: Testing Automated Agents to Assess the Quality of Training Resource Metadata

- The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25)

Towards Foundation Models for Shock Compression of Solid Reactive Materials

  - 2026 International Annual Conference of the Fraunhofer ICT

POSTER PRESENTATIONS

- International Annual Conference of the Fraunhofer ICT - 2026
- Research Days - Kean University 2022 - 2024
- ABRCMS Conference 2024
- Rutgers RISE REU Symposium 2024